

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

6. April 2026

Besi Switzerland AG

Hinterbergstrasse 2

6330 Steinhausen

Switzerland

Bewerbung als Computer Vision & Machine Learning Engineer

Sehr geehrte Damen und Herren,

die Entwicklung hochpräziser Montageautomaten für die Halbleiterindustrie erfordert eine nahtlose Integration modernster Computer-Vision-Algorithmen in performante Steuerungssysteme. Als Physikingenieur mit umfassender Expertise in der Bildverarbeitung, Machine Learning und C++/Python-Entwicklung verfüge ich über das ideale Profil, um Ihr Team bei Besi Switzerland in Steinhausen maßgeblich zu unterstützen. Als gebürtiger Kolumbianer, der seit Jahren im DACH-Raum lebt und arbeitet, kommuniziere ich sicher auf Deutsch und Englisch.

In meiner aktuellen Promotion an der Medizinischen Universität Innsbruck entwickle ich hochkomplexe Phase-Retrieval-Algorithmen und Echtzeitsteuerungssysteme für die adaptive Optik. Ein zentraler Bestandteil meiner Arbeit ist die Implementierung effizienter Bildverarbeitungspipelines in C++ und Python zur Analyse von Wellenfrontfehlern im Sub-Millisekundenbereich. Diese Erfahrung in der Entwicklung robuster Algorithmen für hochpräzise physikalische Systeme bildet die perfekte Grundlage für die Anforderungen bei Besi an die prozessbegleitende Qualitätskontrolle und die hochgenaue Positionierung in Bestückungssystemen.

Dank meiner Zeit bei INTECOL S.A.S. und am Leibniz-IPHT bin ich zudem bestens mit der industriellen Anwendung von **Halcon** (MVTec) und Cognex VisionPro vertraut. Ich habe erfolgreich Inspektionssysteme für Hochgeschwindigkeitsfertigungslinien entwickelt und in die Serienproduktion überführt. Mein fundiertes Verständnis für KI-basierte Bildanalyse, unterstrichen durch mein Python-Paket SpectraMap, ermöglicht es mir, auch komplexe Klassifizierungs- und Segmentierungsaufgaben sicher zu lösen. Meine Leidenschaft für die Perfektionierung technischer Systeme und meine Fähigkeit zur multidisziplinären Zusammenarbeit machen mich zu einem idealen Kandidaten für Ihr Team in der Zentralschweiz.

Ich stehe Ihnen ab dem 1. Juli 2026 zur Verfügung und freue mich sehr darauf, Sie in einem persönlichen Gespräch von meinem Beitrag zur technologischen Marktführerschaft von Besi zu überzeugen.

Mit freundlichen Grüßen,

Juan David Muñoz-Bolaños

Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Computer Vision & Machine Learning Engineer

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+43 676 9115145

juan.munoz@i-med.ac.at | LinkedIn Profile



Summary

Computer Vision & Machine Learning Engineer with deep expertise in **Image Processing, Algorithmic Development, and High-Precision Automation**. Proficient in **Halcon**, C++, and Python for real-time hardware-software integration. Currently finishing a PhD focused on automated wave-field control and system-level optimization for industrial and research applications.

Professional Experience

Mar 2018 to Mar 2019 **INTECOL S.A.S.**

Medellín, Colombia

Junior Machine Vision System Engineer

Machine Vision Pipelines: Developed and commissioned Halcon-based inspection systems for high-speed glass lines (labels, sealing, liquid level). Handled custom illumination design and hardware integration.

Robotic Vision: Implemented Cognex VisionPro for automated part recognition, enabling robotic flame treatment of automotive components (Renault).

Jan 2022 to Present **Medical University of Innsbruck**

Innsbruck, Austria

PhD Researcher, Adaptive Optics & Data Processing

Optics & Metrology: Built a fully automated wavefront sensor stack using C++/Python/LabVIEW; integrated adaptive optics for diffraction-limited imaging, published as "**Confocal Raman Microscopy with Adaptive Optics**" in **ACS Photonics '25**.

System Integration: Full-stack development from hardware interfacing to real-time wave-field control and precision data processing in Python.

Jun 2020 to Dec 2021 **Leibniz IPHT**

Jena, Germany

M.Sc. Researcher | Hyperspectral AI & Photonics

AI & Spectroscopy: Developed the SpectraMap Python package for hyperspectral Raman analysis. Leveraged clustering and ML algorithms for automated cancer detection, published as "**Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer**" in **Applied Sciences '24**.

Spectrometer Prototyping: Led the optical design and build of a 1064 nm Raman spectrometer.

Technical Competencies

Image Processing & Computer Vision: Halcon (MVTec), Cognex VisionPro, OpenCV; object detection, segmentation, defect detection; illumination design (bright-field, dark-field, structured light).

AI & Machine Learning: Supervised ML on image and spectral data (Python, NumPy, scikit-learn); clustering and classification models for automated defect detection and object recognition.

Software Development: Python (advanced) · Halcon (intermediate) · C++ (intermediate / OpenCV) · LabVIEW/FPGA · Git · MATLAB; knowledge of C#/.NET concepts.

3D Metrology & Simulation: Phase retrieval, Zernike decomposition, wave-field modeling in Python; ZEMAX OpticStudio tolerance analysis.

Industrial Integration: Hardware-software integration of camera systems, laser modules, and positioning

stages; high-precision alignment of optical measurement systems; commissioning on production lines.

Laser Systems Alignment: Expertise in integration and alignment of high-power laser systems (fs, CW, 1064 nm); maintenance of precision opto-mechatronic setups and EH&S compliance.

Education

Jan 2022 to Present	Medical University of Innsbruck <i>PhD in Adaptive Optics & Microscopy</i>	Austria
2019 to 2021	Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany
2012 to 2018	University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia

Publications & Awards

Awards: ASML & ZEISS Summer Schools (Selected); COLFUTURO Grant 2026 Awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2 Certificate | High Intermediate) · Spanish (Native)

Soft Skills & Hobbies

Problem Solving: Resolved complex, cross-disciplinary challenges resulting in 3 peer-reviewed publications.

Adaptability: Built a global career path across Colombia, Germany, and Austria with strong networking skills.

Hobbies: Mountain biking, Bachata dancing, harmonica, learning German.

References

Prof. Monika Ritsch-Marte
Head of the Institute of Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft
Spectroscopy Group Leader
christoph.krafft@leibniz-ipht.de

Leibniz-IPHT Jena

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

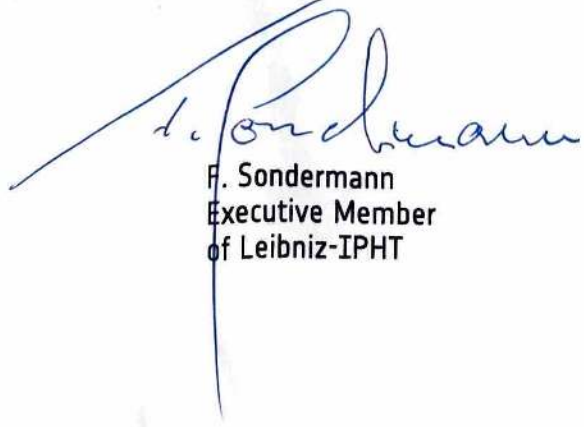
Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

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temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

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Jena, 13 November 2020


Prof. Dr. Jürgen Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

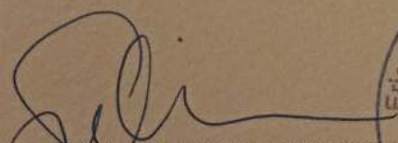
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



ASML

PhD Master Class

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
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Medellin Calle 42 #63c-82
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Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Forma dei tubuli / *Peperidera* e *Aggraderia*

Dear Family

[illegible]